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Research Article

Toughened Bimodal Cathodes for All-Solid-State Batteries via Controlled Interfacial Heterogeneity

Junhee Kang,[∥] Hong Rim Shin,[∥] Jonghyeok Yun, Young Jun Lim, Riyul Kim,* and Jong-Won Lee*

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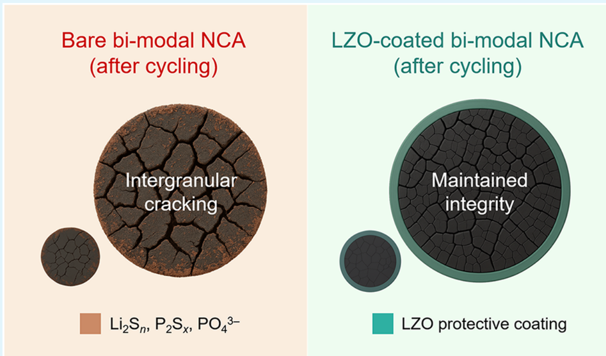
ABSTRACT: All-solid-state batteries (ASSBs) employing composite electrodes require a significant fraction of heavy solid electrolytes (SEs), which poses a challenge to improving their energy density. In this context, a bimodal cathode design, incorporating mixed cathode active materials of different particle sizes, is proposed to enhance the packing density of the cathode layer. This study reveals the mechanism of fracture-induced failure in bimodal cathodes, triggered by interfacial heterogeneity, and discusses the fundamental requirements for achieving high-performance, long-cycling ASSBs. The cycling performance of polycrystalline $LiNi_{0.88}Co_{0.09}Al_{0.03}O_2$ (NCA)|Li_6PS_5Cl (LPSCl)|Li−In full cells is evaluated using two cathode configurations: a unimodal cathode composed solely of 3 μm polycrystalline NCA (U-NCA) and a bimodal cathode comprising a mixture of 3 and 10 μm NCA particles (B-NCA). Although B-NCA offers improved packing density and electronic conductivity, it suffers from rapid capacity decline under external pressures. Comprehensive impedance analysis and microstructural characterization combined with mechanical simulations reveal that the accelerated degradation of B-NCA arises from cracking in the larger NCA particles: the reaction heterogeneity at the NCA/LPSCl interface incurs the grain boundary damage of NCA, which is more severe in larger particles compared to smaller ones. Introducing a Li_2ZrO_3 nanolayer to mitigate the interfacial heterogeneity effectively prevents the mechanical failure of the NCA particles, leading to stable cycling performance with B-NCA. This study provides an in-depth understanding of the degradation characteristics of bimodal cathodes for sulfide-based ASSBs with high energy densities and elucidates the critical factors for their design.



KEYWORDS: *all-solid-state battery, bimodal cathode, high energy density, particle cracking, active material coating*

1. INTRODUCTION

Lithium-ion batteries (LIBs) have played a critical role in the field of energy storage, especially in powering portable electronic devices.[1−3] Their exceptional energy and power densities, coupled with long cycle lives, have solidified their position as the dominant energy storage solution for decades.[4−6] Nonetheless, the continuous pursuit of batteries with higher energy densities and longer lifespans has exposed the inherent limitations of LIBs, including their capacity ceiling and safety concerns arising from flammable liquid electrolytes.[7−10] These issues have encouraged the exploration of next-generation energy storage solutions, where all-solid-state batteries (ASSBs) have emerged as a promising alternative.[11−14] By replacing liquid electrolytes with inorganic solid electrolytes (SEs), ASSBs offer the key advantages of enhanced safety without concerns about fire and explosion.[15−18]

Recently, methods for enhancing the energy density of ASSBs by reducing the thickness of the bulk electrolyte layer or incorporating high-energy-density anode materials, such as Li metal and Si, have been reported.[19−22] As the cathode occupies the largest volume among the cell components, increasing the areal capacity of the cathodes is crucial.[23−25] Unlike LIBs, in which liquid electrolytes facilitate uniform ion transport, ASSB cathodes rely on solid−solid contact between particles.[12,26−28] This structural difference complicates transport of Li^+ to the interior active materials, necessitating the incorporation of SE particles into the composite cathode. Although this integration is essential for forming ionic pathways and electrochemically active interfaces, it inherently limits the fraction of active material in the cathode, introducing significant challenges in achieving high energy densities.[29,30] Thickening the cathode is a direct and intuitive solution for reducing the relative volume of inactive components (e.g., separator and current collector).[31−33] However, thick cathodes often suffer from several drawbacks, including severe rate-capability degradation and

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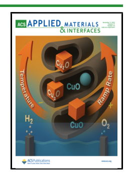

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nonuniform reaction kinetics.[32,33] Therefore, enhancing the packing density by adjusting the particle sizes of the cathode components has been explored using various strategies.[29,34−36] Strauss et al. demonstrated that using smaller active material particles in ASSB cathodes helps decrease the electrochemically inactive portion, thereby enhancing reversible capacity.[36] However, employing small particles requires smaller SE particles to ensure stable interfaces, which not only reduces ionic conductivity in the electrode but also presents challenges in uniformly producing fine-sized SE powders.[37,38]

Consequently, bimodal cathodes combining small and large particles have been explored for ASSB applications, offering the advantage of efficiently filling the gaps between large particles while preserving the electrical connectivity within the cathode.[27,29,39] The maximum achievable active material volume fraction with single-sized particles is limited to 74%, corresponding to the atomic packing factor of a close-packed structure.[40,41] Bimodal cathodes offer the possibility of achieving a higher packing efficiency. In addition to the advantage in terms of the packing density, Han et al. confirmed that utilizing bimodal-sized cathodes enables 98% utilization of the cathode particles by improving the point-to-point contact.[39] Nevertheless, there are numerous factors that must be addressed when integrating bimodal cathodes into ASSBs. For example, high-Ni cathode materials undergo a volume change of approximately 6% during cycling,[42,43] influencing both the mechanical stability of individual particles and the electrode structure, which critically affects the durability of the cell. The mechanical behavior may change considerably, depending on the particle size of the active material in the cathode composite.[44,45] Thorough investigation is required to understand the primary performance-limiting factors in bimodal cathodes, along with considerations and potential solutions for achieving improved performance.

Thus, we comprehensively studied the degradation mechanisms of bimodal cathodes in sulfide-based ASSBs to identify critical strategies for cathode optimization. $LiNi_{0.88}Co_{0.09}Al_{0.03}O_2$ (NCA)|Li_6PS_5Cl (LPSCl)|Li−In full cells were tested using two different cathode setups: a cathode composed solely of 3 μm NCA particles (U-NCA) and a bimodal cathode comprising a mixture of 3 and 10 μm NCA (B-NCA). Electrochemical tests revealed that B-NCA cathodes suffer from poorer rate performance and faster capacity degradation than U-NCA cathodes. In-depth impedance analysis and microstructural characterization confirmed that cracks in the larger particles of B-NCA are the primary cause of the accelerated capacity loss, which was also validated through electrochemical modeling. To minimize these issues, a Li_2ZrO_3 (LZO) coating layer was applied to B-NCA, affording significant improvements in the rate-capability and cycle life. X-ray photoelectron spectroscopy (XPS) and microstructural analyses confirmed that the formation of interfacial byproducts was considerably suppressed in LZO-coated B-NCA compared to that in bare B-NCA, accompanied by a marked reduction in the mechanical degradation of the larger active material particles. Through electrochemo-mechanical simulations, this study demonstrates that enhancing the surface homogeneity of NCA through surface coating effectively mitigates both the Li concentration gradient and mechanical stress in large NCA particles, thus reducing grain boundary damage. Our findings highlight the necessity of mitigating the cracking of the large active material particles for effective utilization of bimodal cathode materials and demonstrate that such degradation can be addressed by introducing a passivating surface layer that can reduce the interfacial heterogeneity.

2. EXPERIMENTAL SECTION

2.1. Electrode Preparation. Polycrystalline NCA and LPSCl powders were supplied by Samsung SDI Co., Ltd., Republic of Korea. The composite cathodes were prepared by ball-milling a blend of NCA, LPSCl, and carbon black (Super P) in a 72:27:1 weight ratio. The Li−In counter electrodes were prepared by layering Li (thickness = 200 μm and diameter = 4 mm) and In (thickness = 100 μm and diameter = 10 mm) foils. LZO was coated on NCA using the literature method.[46]

2.2. Assembly of ASSBs. ASSBs were fabricated using stainless steel rods with a diameter of 10 mm and a poly(aryl-ether-ether−ketone) mold according to the following steps: Initially, 0.1 g of LPSCl powder was placed in a mold and compressed into pellets to form a separator layer. The prepared composite cathode was then evenly distributed on one side of the SE layer and subjected to a pressure of 400 MPa. In and Li foils were attached to opposite sides of the SE layer, and stainless steel current collectors were placed on both sides. The fabricated cells were cycled at a stack pressure of 200 MPa. All the experimental procedures were conducted in an Ar-filled glovebox to avoid exposure to humid air.

2.3. Electrochemical Measurements. Cycling tests were performed at 30 °C in the voltage range of 2.5−4.3 V vs Li/Li$^+$. For the initial cycle, a 0.1C rate was applied, using constant current-constant voltage (CC−CV) mode for charging and CC mode for discharging. The cutoff current in the CV mode was adjusted to 20% of the applied current in the CC mode. Electrochemical impedance spectroscopy (EIS) measurements were performed in the frequency range of 7 MHz to 10 mHz, with an AC amplitude of 5 mV (Biologic SP-300).

2.4. Material Characterization. A scanning electron microscopy (SEM, Verios G4UC, FEI company) analysis was used to examine the morphologies and microstructures of the powders and the cross-sectional surfaces of the composite electrodes. Elemental mapping was simultaneously performed by using energy-dispersive X-ray spectroscopy (EDS). The chemical compositions were analyzed by XPS using focused monochromatized Al K$_\alpha$ radiation ($h\nu$ = 1486.6 eV) by employing an ESCALAB 250Xi instrument (Thermo Scientific).

2.5. Modeling. Finite element analysis was used to investigate the stresses caused by volume changes in the active materials. A polycrystalline NCA structure was created via the Voronoi tesselation method with the secondary particle diameter set to 3 and 10 μm. The cathode domain was defined as 12.5 μm in the x- and y- directions, considering the NCA and LPSCl volume fractions in the actual electrode. The elastic moduli of NCA and LPSCl were assumed to be 175 and 22.1 GPa, respectively.[42] The diffusion-induced strain (ε_d) in NCA due to lithiation/delithiation was calculated using $\varepsilon_d = \Omega/3 \times \Delta c_{Li}$, where Ω is the partial molar volume derived from a 5.9% volume change in NCA, and c_{Li} represents the Li concentration. Based on the cohesive zone model, the extent of grain boundary decohesion was quantified using a scalar variable, called "damage", which ranges from 0 (intact interface) to 1 (fully separated interface). As we assumed that the stress−strain curve during the debonding process is described by a bilinear traction-separation law, the value of damage is expressed by eq 1

$$\text{damage} = \begin{cases} 0 & d_{m,max} < d_{m0} \\ \min\left[\dfrac{d_{fm}}{d_{m,max}}\left(\dfrac{d_{m,max} - d_{m0}}{d_{fm} - d_{m0}}\right), 1\right] & d_{m,max} > d_{m0} \end{cases} \quad (1)$$

where $d_{m,max}$ and d_{fm} represent the maximum value of mixed-mode displacement and the point of complete fracture, respectively. d_{m0} is the critical mixed-mode displacement at which damage initiates. Additional modeling details are presented in Supporting Note 1.

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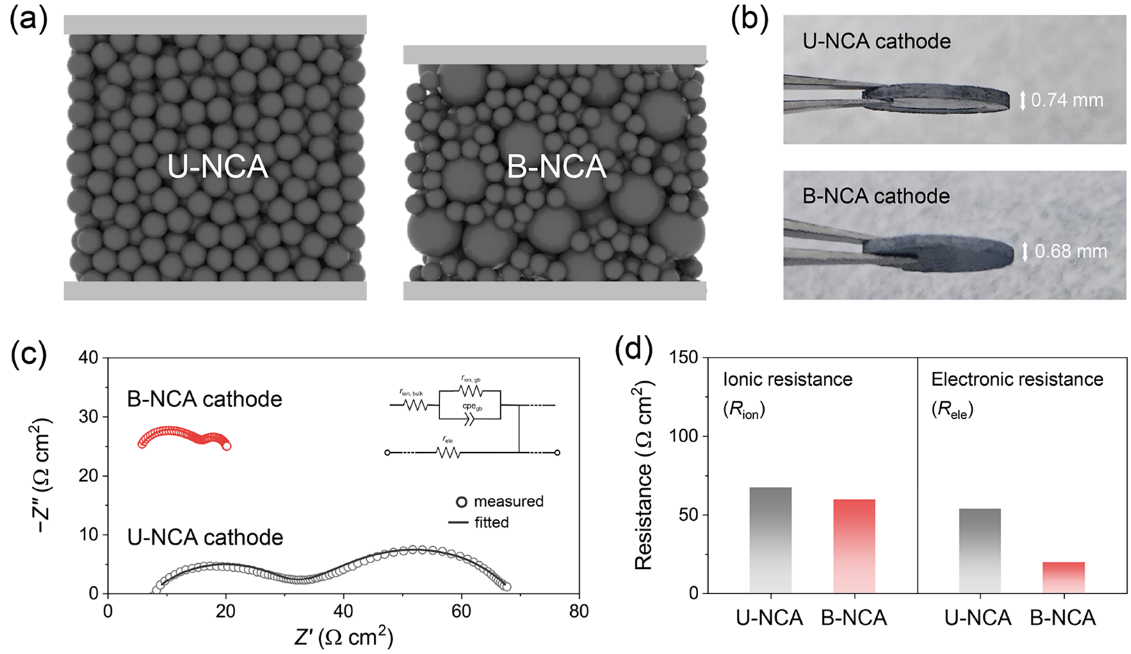

Figure 1. (a) Schematic of stacking structure of active materials in the composite cathode depending on particle size distribution. (b) Photographs of pellets made by compressing 0.15 g of composite cathodes with U-NCA and B-NCA at 400 MPa. (c) Nyquist plots of cathode symmetric cells. (d) Quantified ionic and electronic resistances of U-NCA and B-NCA cathodes.

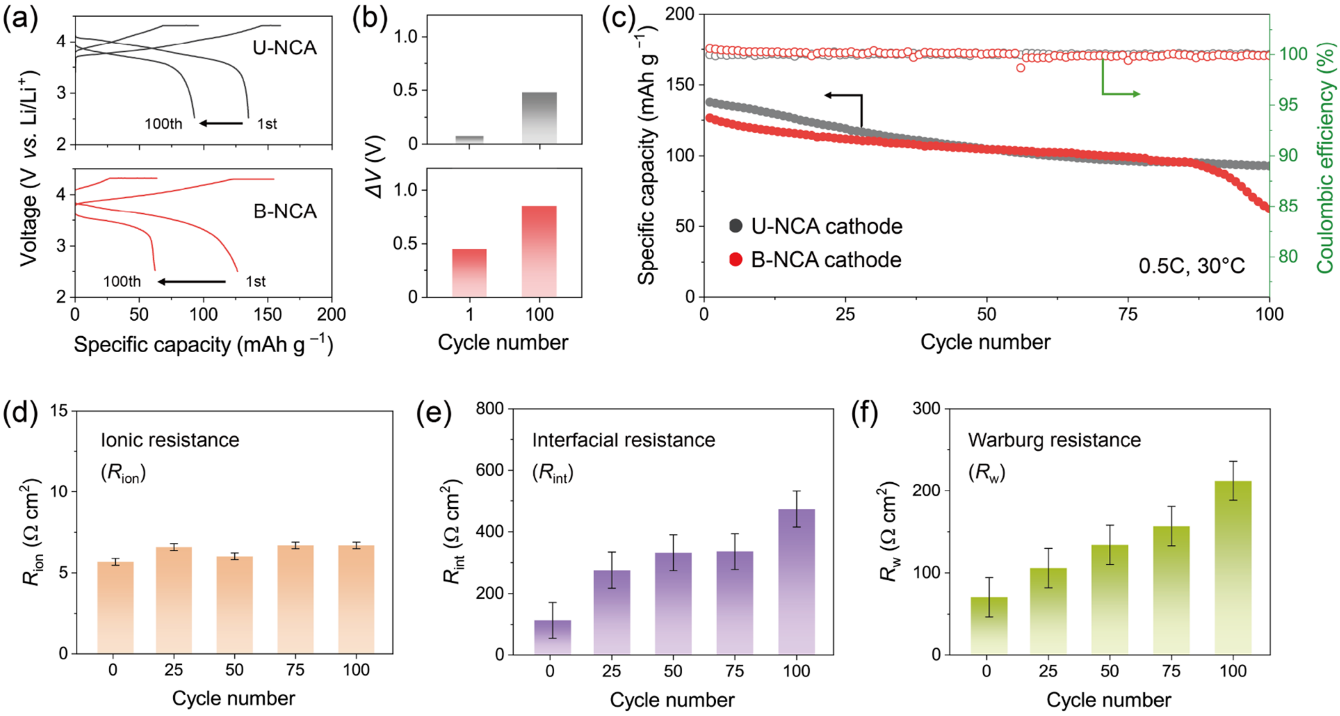

Figure 2. (a) Voltage profiles, (b) charge−discharge voltage gaps at SOC = 50%, and (c) cycle performances of NCA|LPSCl|Li−In cells with U-NCA and B-NCA cycled at 0.5C. (d) Ionic, (e) interfacial, and (f) Warburg resistances (Ω cm^2) of B-NCA cells measured every 25 cycles.

3. RESULTS AND DISCUSSION

Figure 1a shows a composite cathode composed solely of small polycrystalline NCA (U-NCA), and the bimodal cathode (B-NCA) with a blend of small and large particles. As depicted in the schematic, the bimodal cathode offers enhanced stacking efficiency while preserving the electrical connectivity among the active material particles. Two types of active materials with particle sizes of 3 and 10 μm (Figure S1) were employed to examine the effects of dual-sized cathode active materials on the energy density of the cathodes. Figure 1b shows the 10 mm-diameter pellets made from the composite cathodes (0.15 g), which were used to examine how bimodal-sized cathode materials affect the packing density of electrodes. The measured thicknesses were 0.74 mm for U-NCA and 0.68 mm for B-NCA. The volumetric loading (mg cm^{-3}) of NCA in B-NCA was calculated to be 1.1 times higher than that in U-

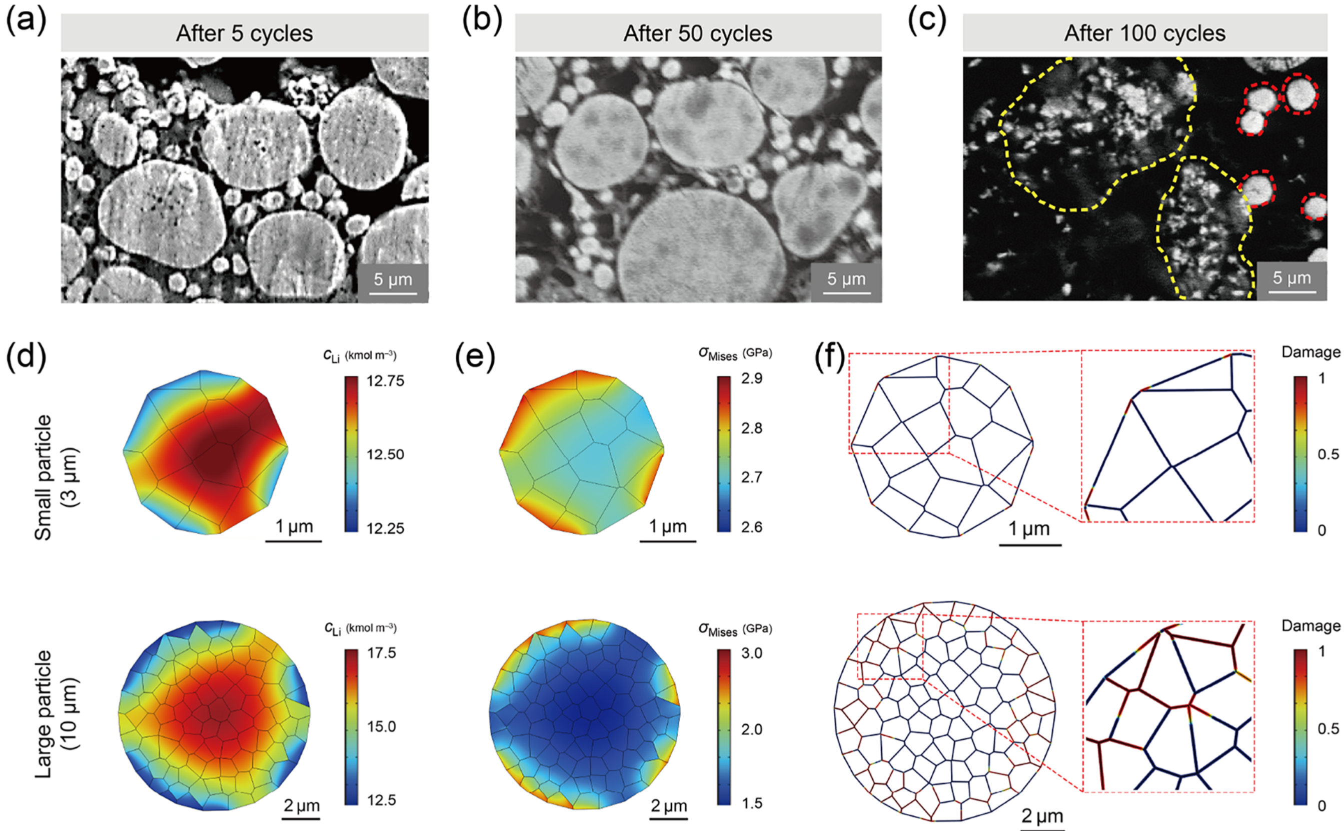

Figure 3. Cross-sectional SEM images of the bare B-NCA cathodes collected after (a) 5, (b) 50, and (c) 100 cycles. Distributions of (d) Li concentration and (e) von Mises stress simulated upon charging at 0.5C with respect to the particle size of active materials. Heterogeneous surface features arising from oxidative decomposition in the cathode were reflected in the simulation. (f) Damage at boundaries between primary particles, indicating intergranular separation inside the secondary particles. A damage value of unity indicates complete detachment.

NCA under the present experimental conditions (Figure S2). This confirms that employing bimodal cathode active materials can significantly enhance the energy density of the cathode.

Figure 1c,d presents the Nyquist plots and quantified ionic and electronic resistances of the cathode symmetric cells, derived by fitting the impedance data to the equivalent circuit in Figure 1c. The ionic resistances (R_{ion}) of U-NCA and B-NCA were similar at 67.4 and 59.8 Ω cm^2, respectively, whereas the electronic resistance (R_{ele}) of B-NCA was 33.9 Ω cm^2 lower than that of U-NCA (Figure 1d and Table S1). For U-NCA, the active materials are electrically isolated by SE particles of a similar size, whereas for B-NCA, a percolated electron pathway is observed to be well maintained through the larger NCA particles. To further evaluate the performance of full cells employing each cathode configuration, NCA|LPSCl|Li−In cells were tested at a 0.5C rate (Figure 2a−c). U-NCA showed a slightly higher initial discharge capacity (183.1 mAh g^{-1} for U-NCA and 177.1 mAh g^{-1} for B-NCA) as well as superior rate performance than B-NCA (Figure S3). The cathode composed solely of large NCA particles delivered a lower discharge capacity, mainly due to the reduced contact area between the active material and SE (Figure S4). Moreover, although U-NCA maintained stable cycling for over 100 cycles (capacity retention = 67.3%), B-NCA experienced rapid capacity decay after 80 cycles, with a retention of only 47.7% (Figure 2c) under a high external pressure of 200 MPa. Evaluation of the voltage gap (ΔV) at a state of charge (SOC) = 50% demonstrates a significant increase in ΔV for the B-NCA cell relative to that of the U-NCA cell (Figure 2b). Thus, although B-NCA offers advantages in terms of stacking efficiency and electronic conductivity, its electrochemical performance is inferior to that of U-NCA.

To identify the cause of the steep capacity decline observed for B-NCA, the resistance within the composite cathodes during cycling was quantified through an *in situ* impedance analysis. Impedance measurements were taken every 25 cycles throughout the 100-cycle operation (Figure S5), and the Nyquist plots were fitted to the equivalent circuit based on the transmission-line model (TLM) (Figure S6). TLM represents the distributed nature of ion/electron transport and interfacial reactions across the electrode thickness, offering a realistic description of porous composite cathodes with intertwined ionic and electronic networks.[47] The extracted parameters are listed in Table S2. The resistance at the anode interface (R_{anode}) remained nearly constant after 100 cycles (Figure S7), whereas the cathode resistance increased significantly during operation. Figure 2d−f illustrates the evolution of three major resistance values that contribute to cathode degradation: ionic resistance (R_{ion}), interfacial resistance (R_{int}), and Warburg resistance (R_w). R_{ion} reflects the extent to which Li^+ transport is hindered within the cathode, and R_{int} represents the degree of interfacial degradation and contact loss at the NCA/LPSCl interface. R_w measures the kinetics of diffusion in the cathode materials, which is described by eq 2

$$R_w = \frac{RT\delta_s}{z^2F^2c^*_{Li^+,s}D_s} \quad (2)$$

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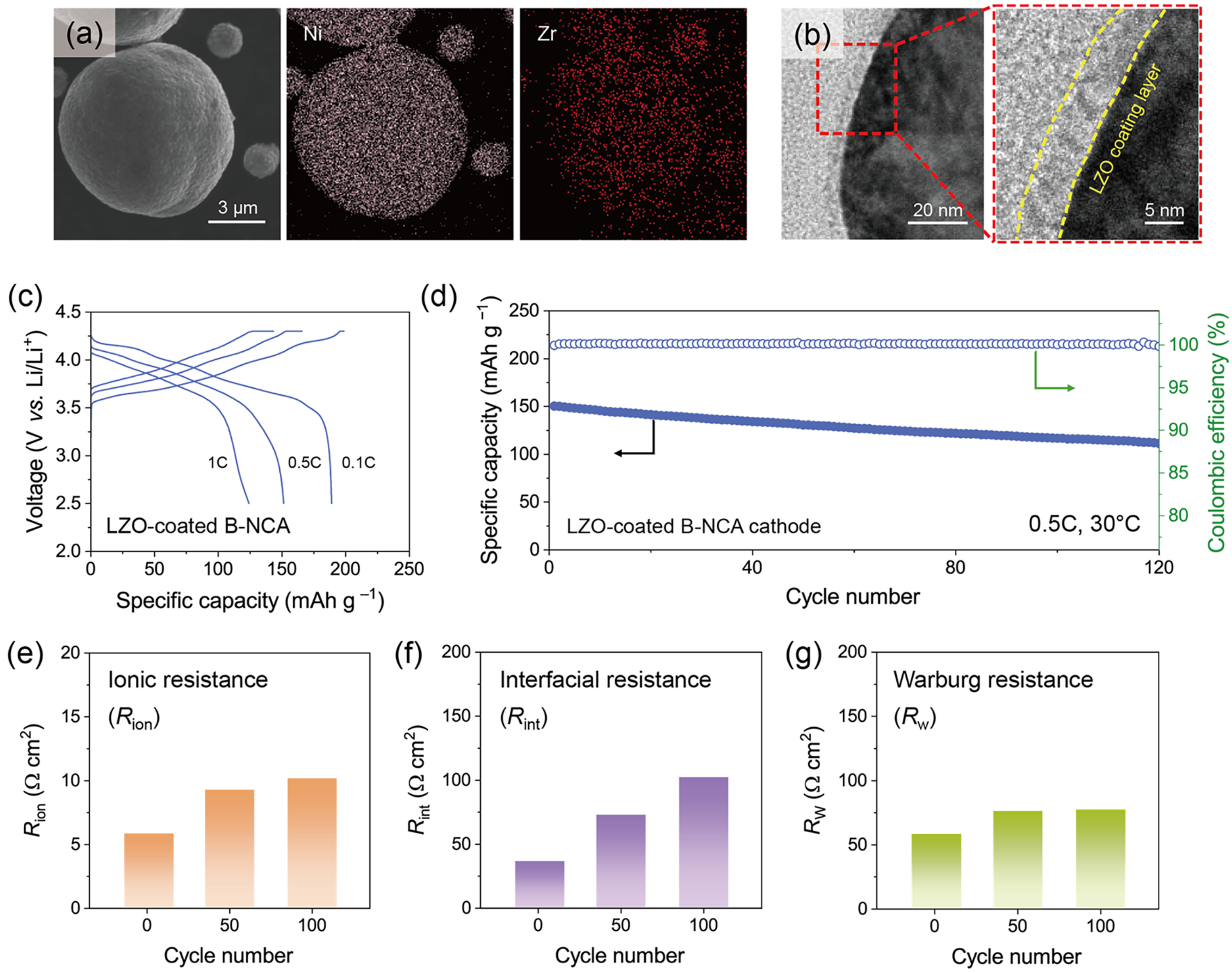

Figure 4. (a) SEM and EDS images of LZO-coated B-NCA particles. (b) Cross-sectional TEM images of LZO-coated B-NCA. (c) Voltage profile of the ASSB with LZO-coated B-NCA cathode during the rate-capability test from 0.1 to 1.0C. (d) Capacity retention and Coulombic efficiency of the ASSB with LZO-coated B-NCA cathode. (e) Ionic, (f) interfacial, and (g) Warburg resistances (Ω cm^2) of cells with U-NCA and B-NCA cathodes measured every 50 cycles.

where z is the number of electrons consumed, F is the Faraday constant, $c_{Li^+,s}$ is the bulk concentration of Li^+, D_s is the effective diffusion coefficient for Li^+, R is the gas constant, T is the temperature, and δ_s is the diffusion distance.[47] While R_{ion} showed only a minimal increase from 5.7 to 6.7 Ω cm^2, both R_{int} and R_w experienced a sharp rise, particularly between the 75th and 100th cycles. These findings suggest that the ion transport pathways within the composite cathode remained relatively intact, but significant interfacial degradation, likely stemming from contact loss and SE decomposition, occurred, coupled with an increase in the tortuosity of the diffusion pathway in NCA. This difference is attributed to the large NCA particles in B-NCA, which reduced the contact area with the SE and provided long diffusion pathways, leading to less favorable performance at high rates. The U-NCA cathode maintained stable R_{int} and R_w behavior without abrupt resistance growth after 100 cycles, indicating that severe mechanical degradation occurs specifically in the B-NCA cathode (Figure S8 and Table S3). After the first cycle, the R_w values of the U-NCA and B-NCA cathodes were measured to be 59.5 and 70.7 Ω cm^2, respectively. Considering the proportional relationship between R_w and δ_s (eq 2), these results indicate that the inferior rate performance of B-NCA arises from its longer diffusion distance.

Our earlier electrochemical analyses revealed a distinct shift in the degradation behavior of B-NCA after 80 cycles, which is potentially related to mechanical failure (reduced structural integrity) of the active materials within B-NCA. SEM analysis was carried out on cross-sectional samples of the composite cathode obtained after 5, 50, and 100 cycles to examine microstructural changes during different degradation stages. No particle fracture was observed after a fabrication pressure of 400 MPa was applied (Figure S9). After 5 and 50 cycles, the structural integrity of the small and large particles in the B-NCA cathode was maintained (Figure 3a,b). However, by the 100th cycle, severe cracking was observed, predominantly in the large polycrystalline particles (Figures 3c and S10). Cathode active materials, particularly those of the high-Ni cathodes used in this experiment, are prone to mechanical stress due to significant volume changes during repeated charge and discharge.[42,43] ASSBs inherently depend on solid−solid contact, which gives rise to nonuniform contact at NCA/

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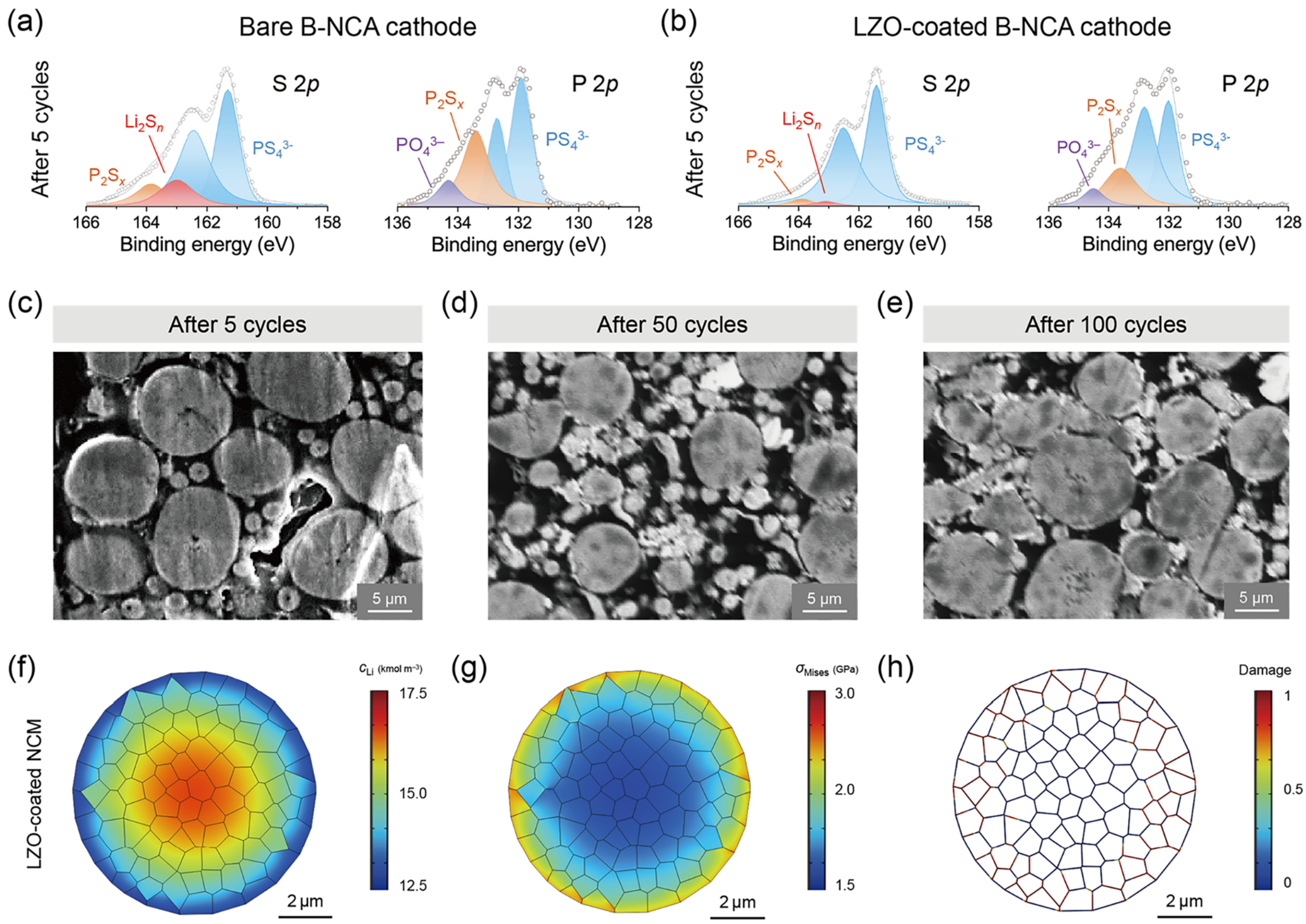

Figure 5. XPS spectra for S 2p and P 2p core levels after 5 cycles of composite cathodes with (a) bare B-NCA and (b) LZO-coated B-NCA. Cross-sectional SEM images of LZO-coated B-NCA cathodes after (c) 5, (d) 50, and (e) 100 cycles at 0.5C. Distribution of (f) Li concentration and (g) von Mises stress in LZO-coated NCA upon charging at 0.5C. (h) Damage at the boundaries between primary particles in large LZO-coated NCA at SOC = 100%, showing intergranular separation inside the secondary particles. A damage value of unity signifies complete detachment.

LPSCl interfaces and consequently initiates localized side reactions. The resulting surface heterogeneity of active materials, coupled with the repetitive volume change during cycling, generates uneven stress distributions that accelerate mechanical degradation and eventually leads to particle cracking.[27] These cracks are reported to be strongly associated with a loss of reversible capacity in ASSBs. The results in Figure 2c demonstrate that this phenomenon occurs more prominently in large active material particles and may serve as a major factor limiting the lifespan of bimodal cathodes.

We employed electrochemical-mechanical modeling to explore the relationship among the surface heterogeneity, active material size, and mechanical stress. As in the experiments, the particle sizes of NCA were set to 3 and 10 μm, and the electrochemo-mechanical behavior of NCA particles at SOC = 100% was monitored in the SE domain with an elastic modulus of 22.1 GPa. This simulation accounted for the surface heterogeneity induced by interfacial reactions in the cathode in addition to the inherent material properties and electrode-specific parameters. The detailed parameters and boundary conditions for modeling are provided in Tables S4 and S5. A 10-fold increase in the c_{Li} gradient was identified in the large particles relative to the small ones (Figure 3d), indicating that the long diffusion pathway caused significant spatial inhomogeneity in c_{Li}. Such intraparticle c_{Li} gradients can cause cracking of polycrystalline materials.[48,49] This nonuniform Li distribution in NCA induced a significant von Mises stress (σ_{Mises}) gradient at the surface of the large particles, as shown in Figure 3e. σ_{Mises}, a parameter commonly used to predict material yielding, serves as a critical indicator for assessing the mechanical degradation of cathode materials.[50,51] A high stack pressure of 200 MPa applied in our experiments may have a certain degree of influence, since excessive stack pressure is known to elevate the internal stress of particles, thereby inducing plastic deformation and particle fracture.[52] However, the stack pressure-induced stress, typically on the order of several hundreds of MPa, is much smaller than the diffusion-induced stress, which can reach several GPa.[27] Therefore, the contribution of stack pressure to particle cracking is negligible compared to that arising from the c_{Li} inhomogeneity. This finding highlights that the primary origin of capacity fade in B-NCA cathodes is the fracture of larger particles. The damage value, which quantifies the grain boundary separation between primary particles, approached 1 at multiple boundaries in the 10 μm particles, indicative of complete grain boundary detachment (Figure 3f).

To reduce the interfacial heterogeneity at the NCA/LPSCl interface and avoid chemo-mechanical cracking of the cathode material, LZO-coated NCA was synthesized using the sol−gel method. LZO was selected as the coating material because it

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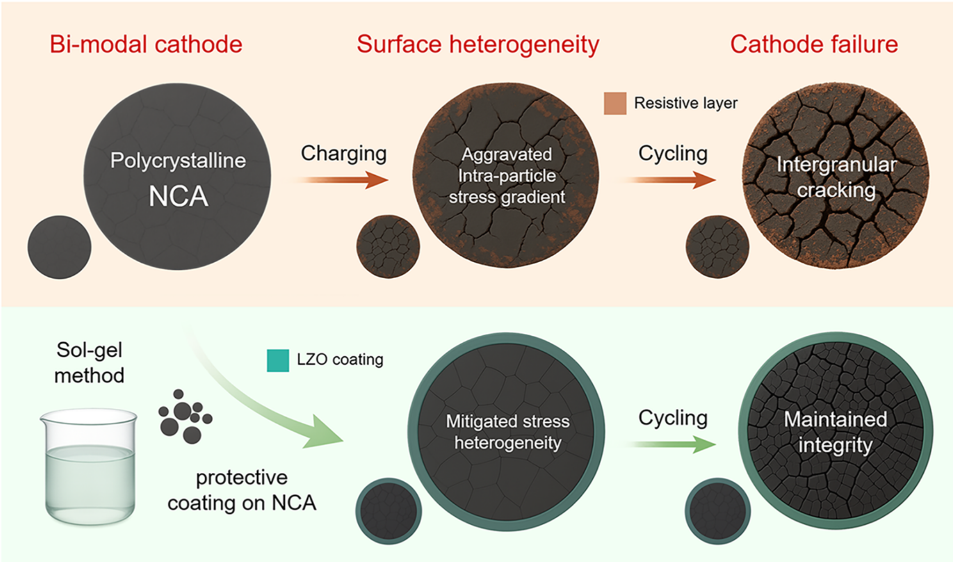

Figure 6. Schematic illustration of mechanical degradation in composite cathodes of sulfide-based ASSBs with bimodal cathodes and its mitigation by applying LZO coating layer.

exhibits superior chemical stability against sulfide SEs, while possessing a higher ionic conductivity than conventional oxides such as $LiNbO_3$ and $Li_4Ti_5O_{12}$, thereby sustaining efficient ionic pathways within the cathode.[46] The LZO coating layer exists in an amorphous phase with no discernible lattice fringes and was uniformly deposited with a thickness of 6−8 nm (Figure 4a,b). The microstructure of the B-NCA electrode showed no significant changes before and after coating (Figure S11). A composite cathode (0.15 g) fabricated with LZO-coated B-NCA exhibited a thickness of 0.67 mm, nearly identical to that of bare B-NCA (0.68 mm), indicating that the nanoscale LZO coating has a negligible influence on the overall energy density of cathodes. As presented in Figure 4c,d, significant improvements were observed in both rate-capability and cycling performance, with stable operation maintaining over 120 cycles at a 0.5C rate. The LZO coating on B-NCA contributed to a 54.4 mAh g^{-1} increase in discharge capacity at the 100th cycle compared to that of the uncoated counterpart. Quantitative impedance analysis at 1, 50, and 100 cycles showed a notable decrease in R_{int} and R_w at the 100th cycle compared to those of bare B-NCA (Figure 4e−g). The fitting parameters are summarized in Table S6. Although the reduction of interfacial contact area due to the volume change of high-strain NCA made a partial contribution to the gradual increase of R_{int} during cycling, the LZO-coated B-NCA cathode exhibited a significantly reduced R_{int} compared to the bare B-NCA cathode (Figure 2e) throughout cycling. This indicates that the LZO layer effectively mitigated byproduct formation at the SE/NCA interface. No sharp increase of R_{int} and R_w was observed for the LZO-coated B-NCA cathodes; notably, R_w increased by only 1.2 Ω cm^2 between the 50th and 100th cycles, indicating that R_w is directly linked to the lifespan of bimodal cathodes.

Further analyses were conducted to determine the effectiveness of the LZO coating on B-NCA. Figure 5a,b compares the XPS spectra of the S 2p and P 2p core levels in the composite cathodes collected after 5 cycles for bare B-NCA and LZO-coated B-NCA. Each spectrum was collected after charging the cell to 4.3 V vs Li/Li^+. The S 2p spectra peaks at 161.3 and 162.5 eV are attributed to the characteristic PS_4^{3-} tetrahedra of LPSCl.[53−55] The XPS profiles obtained after 5 cycles demonstrate that the LZO coating successfully reduced degradation at the NCA/LPSCl interface during the early phase of cycling. P_2S_x (133.5 eV), Li_2S_n (163.0 eV), and PO_4^{3-} (134.4 eV) are byproducts generated from the oxidative decomposition of LPSCl, indicating that B-NCA experienced significant degradation at the NCA/LPSCl interface during cycling. It can be inferred that the LZO layer served as an efficient passivation barrier against undesirable interfacial reactions at the NCA/LPSCl interface.

Given that the cracking of 10 μm particles was identified as the primary cause of capacity fading in bare B-NCA, postcycling microstructural analysis was conducted for the LZO-coated B-NCA for comparison (Figure 5c−e). As shown earlier in Figure 2c, severe mechanical cracking occurred in large active material particles in bare B-NCA. In contrast, after applying the LZO coating, not only the small particles but also the large particles maintained a high level of integrity even after 100 cycles (Figure 5e). This confirms that the cracking of large active material particles, a major issue with B-NCA, can be effectively controlled by the coating layer. To investigate how coating-induced passivation, as confirmed by XPS analysis, affects the mechanical degradation of active materials, simulations were performed on large NCA particles with improved surface homogeneity (Figure 5f−h). The suppression of interfacial side reactions by the LZO coating led to more uniform electrochemical reactions across the particle surface, thereby considerably alleviating both the Li concentration and σ_{Mises} gradients compared to those of the uncoated sample. Accordingly, a notable reduction in the intergranular detachment between the primary particles was observed compared with that in Figure 3f (Figure 5h).

Figure 6 provides a schematic of the mechanical degradation in the large active material particles within the bimodal cathodes of sulfide-based ASSBs, and how LZO coating alleviates this degradation by enhancing surface uniformity and managing volume changes during Li intercalation/deintercalation. For bare B-NCA, the oxidation of LPSCl at high voltages causes significant surface heterogeneity, resulting in substantial stress gradients in the larger particles of the composite. This

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induces severe chemo-mechanical failure in large particles, ultimately resulting in rapid capacity fading. The application of the LZO coating on B-NCA via a sol−gel process results in a notable improvement in the surface uniformity of NCA, enabling significantly enhanced cell durability.

4. CONCLUSIONS

In summary, we performed extensive analyses to investigate the electrochemo-mechanical degradation mechanisms of bimodal cathodes in sulfide-based ASSBs, and proposed a surface engineering strategy to address the limitations posed by large active material particles. While the incorporation of large particles improved the packing density and electronic conductivity of the cathode, the B-NCA cathode displayed a relatively lower initial Coulombic efficiency and shorter cycle life compared to the U-NCA cathode. Electrochemical analyses, microstructural observations, and modeling collectively revealed that the primary cause of capacity degradation in bare B-NCA is the cracking of 10 μm-sized particles, initiated by spatially inhomogeneous Li concentration and stress gradients due to the decomposition at NCA/LPSCl interfaces. To mitigate this chemo-mechanical degradation, a conformal LZO coating was introduced via a sol−gel method. The LZO layer effectively passivated the NCA/LPSCl interface by suppressing interfacial side reactions and notably improved the surface uniformity of NCA. As a result, the LZO-coated B-NCA demonstrated enhanced mechanical integrity, leading to significantly improved capacity retention over extended cycling. These findings underscore the importance of interfacial engineering in preserving the structural and electrochemical stability of high-energy-density cathodes in ASSBs. The strategy of applying a passivating ceramic coating to mitigate localized degradation offers practical design insights for the implementation of bimodal cathodes in sulfide-based ASSBs. Considering our finding that the dominant degradation in bimodal cathodes originates from Li concentration and stress gradients within the large cathode particles, there are several other potential strategies that can further improve the cyclability of bimodal cathodes. Optimizing the particle size and fraction of the cathode particles can alleviate mechanical failure of composite cathodes, and the surface doping of cathode materials offers an additional route to stability transition-metal sites or induces the formation of protective films.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.5c14519.

> Detailed information on electrochemical modelig principles, SEM images of small and large NCA particles in B-NCA, volumetric loading of composite cathodes with U-NCA and B-NCA, fitted parameters of the EIS spectra obtained from cathode symmetric cells, rate performance tests of NCA|LPSCl|Li−In cells with U-NCA and B-NCA, AC impedance spectra of B-NCA|LPSCl|Li−In cells measured every 25 cycles at SOC = 100%, equivalent circuit model used to fit the Nyquist plots of NCA|LPSCl|Li−In full cells, fitted parameters of the EIS spectra obtained from B-NCA|LPSCl|Li−In full cells during cycling at 0.5C, quantified R_{anode} values the of B-NCA|LPSCl|Li−In cell measured every 25 cycles, AC impedance spectra of U-NCA|LPSCl|Li−In cells, cross-sectional SEM images of the composite cathodes after cell fabrication, cross-sectional SEM images of the cracked large NCA particles in the B-NCA cathode collected after 100 cycles, parameters for electrochemical modeling, boundary conditions for electrochemical modeling, TEM images of LZO-coated B-NCA, cross-sectional SEM images of the composite cathodes employing bare B-NCA and LZO-coated B-NCA active materials, and fitted parameters of the EIS spectra obtained from LZO-coated B-NCA|LPSCl|Li−In full cells during cycling at 0.5C (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Riyul Kim − *Samsung SDI Co., Ltd., Suwon-si, Gyeonggi-do 433-803, Republic of Korea*; Email: riyul.kim@samsung.com

Jong-Won Lee − *Division of Materials Science and Engineering, Hanyang University, Seoul 04763, Republic of Korea; Department of Battery Engineering, Hanyang University, Seoul 04763, Republic of Korea*; orcid.org/0000-0002-8713-7143; Email: jongwonlee@hanyang.ac.kr

Authors

Junhee Kang − *Division of Materials Science and Engineering, Hanyang University, Seoul 04763, Republic of Korea*

Hong Rim Shin − *Division of Materials Science and Engineering, Hanyang University, Seoul 04763, Republic of Korea*

Jonghyeok Yun − *Division of Materials Science and Engineering, Hanyang University, Seoul 04763, Republic of Korea*

Young Jun Lim − *Samsung SDI Co., Ltd., Suwon-si, Gyeonggi-do 433-803, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.5c14519

Author Contributions

[‖]J.K. and H.R.S. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Wang, Z.; Du, Z.; Wang, L.; He, G.; Parkin, I. P.; Zhang, Y.; Yue, Y. Disordered Materials for High-Performance Lithium-Ion Batteries: A Review. *Nano Energy* **2024**, *121*, No. 109250.

(2) Wang, R.; Wang, L.; Liu, R.; Li, X.; Wu, Y.; Ran, F. "Fast-Charging" Anode Materials for Lithium-Ion Batteries from Perspective of Ion Diffusion in Crystal Structure. *ACS Nano* **2024**, *18* (4), 2611−2648.

(3) Jia, K.; Yang, G.; He, Y.; Cao, Z.; Gao, J.; Zhao, H.; Piao, Z.; Wang, J.; Abdelkader, A. M.; Liang, Z.; Kumar, R. V.; Zhou, G.; Ding, S.; Xi, K. Degradation Mechanisms of Electrodes Promotes Direct Regeneration of Spent Li-Ion Batteries: A Review. *Adv. Mater.* **2024**, *36* (23), No. 2313273.

(4) Chen, J.; Wu, J.; Wang, X.; Zhou, A.; Yang, Z. Research Progress and Application Prospect of Solid-State Electrolytes in Commercial Lithium-Ion Power Batteries. *Energy Storage Mater.* **2021**, *35*, 70−87.

(5) Cao, D.; Zhao, Y.; Sun, X.; Natan, A.; Wang, Y.; Yu, X.; Zhu, Y.; Xin, S.; Guo, Y.; Zhang, X.; Cui, Y.; Lin, D. Processing Strategies to Improve Cell-Level Energy Density of Metal Sulfide Electrolyte-Based All-Solid-State Li Metal Batteries and Beyond. *ACS Energy Lett.* **2020**, *5* (11), 3468−3489.

(6) Gicha, B. B.; Tufa, L. T.; Nwaji, N.; Hu, X.; Lee, J. Advances in All-Solid-State Lithium-Sulfur Batteries for Commercialization. *Nano-Micro Lett.* **2024**, *16*, No. 172.

(7) Hu, X.; Gao, F.; Xiao, Y.; Wang, D.; Gao, Z.; Huang, Z.; Ren, S.; Jiang, N.; Wu, S. Advancements in the Safety of Lithium-Ion Battery: The Trigger, Consequence and Mitigation Method of Thermal Runaway. *Chem. Eng. J.* **2024**, *481*, No. 148450.

(8) Liu, M.; Zeng, Z.; Wu, Y.; Zhong, W.; Lei, S.; Cheng, S.; Wen, J.; Xie, J. Reviewing Recent Progress of Liquid Electrolyte Chemistry for Mitigating Thermal Runaway in Lithium-Ion Batteries. *Energy Storage Mater.* **2024**, *65*, No. 103133.

(9) Williams, D.; Copley, R.; Bugryniec, P.; Dwyer-Joyce, R.; Brown, S. A Review of Ultrasonic Monitoring: Assessing Current Approaches to Li-Ion Battery Monitoring and Their Relevance to Thermal Runaway. *J. Power Sources* **2024**, *590*, No. 233777.

(10) Peng, R.; Kong, D.; Ping, P.; Wang, G.; Gao, X.; Lv, H.; Zhao, H.; He, X.; Zhang, Y.; Dai, X. Thermal Runaway Modeling of Lithium-Ion Batteries at Different Scales: Recent Advances and Perspectives. *Energy Storage Mater.* **2024**, *69*, No. 103417.

(11) Huang, J.; Li, C.; Jiang, D.; Gao, J.; Cheng, L.; Li, G.; Luo, H.; Xu, Z. L.; Shin, D. M.; Wang, Y.; Lu, Y.; Kim, Y. Solid-State Electrolytes for Lithium Metal Batteries: State-of-the-Art and Perspectives. *Adv. Funct. Mater.* **2024**, *34* (1), No. 2411171.

(12) Zhang, Q.; Cao, D.; Ma, Y.; Natan, A.; Aurora, P.; Zhu, H. Sulfide-Based Solid-State Electrolytes: Synthesis, Stability, and Potential for All-Solid-State Batteries. *Adv. Mater.* **2019**, *31* (44), No. 1901131.

(13) Cho, M.; Yun, J.; Kang, J.; Kim, S.; Lee, J.-W. Conflicting Roles of Conductive Additives in Controlling Cathode Performance in All-Solid-State Batteries. *Electrochim. Acta* **2024**, *481*, No. 143990.

(14) Woolley, H. M.; Vargas-Barbosa, N. M. Hybrid Solid Electrolyte−Liquid Electrolyte Systems for (Almost) Solid-State Batteries: Why, How, and Where To? *J. Mater. Chem. A* **2023**, *11* (3), 1083−1097.

(15) Tan, D. H. S.; Meng, Y. S.; Jang, J. Scaling Up High-Energy-Density Sulfidic Solid-State Batteries: A Lab-to-Pilot Perspective. *Joule* **2022**, *6* (8), 1755−1769.

(16) Kong, W. J.; Zhao, C. Z.; Sun, S.; Shen, L.; Huang, X. Y.; Xu, P.; Lu, Y.; Huang, W. Z.; Huang, J. Q.; Zhang, Q. From Liquid to Solid-State Batteries: Li-Rich Mn-Based Layered Oxides as Emerging Cathodes with High Energy Density. *Adv. Mater.* **2024**, *36* (14), No. 2310738.

(17) Cha, H.; Yun, J.; Kim, S.; Kang, J.; Cho, M.; Cho, W.; Lee, J.-W. Stabilizing the Interface Between High-Ni Oxide Cathode and Li_6PS_5Cl for All-Solid-State Batteries via Dual-Compatible Halides. *J. Power Sources* **2024**, *617*, No. 235157.

(18) Du, H.; Wang, Y.; Kang, Y.; Zhao, Y.; Tian, Y.; Wang, X.; Tan, Y.; Liang, Z.; Wozny, J.; Li, T.; Ren, D.; Wang, L.; He, X.; Xiao, P.; Mao, E.; Tavajohi, N.; Kang, F.; Li, B. Side Reactions/Changes in Lithium-Ion Batteries: Mechanisms and Strategies for Creating Safer and Better Batteries. *Adv. Mater.* **2024**, *36* (29), No. 2401482.

(19) Lei, Y.; Wang, K.; Jiang, S.; Xu, X.; Zheng, J.; Yin, J.; Gao, Y. Recent Progress on Multifunctional Electrolyte Additives for High-Energy-Density Li Batteries − A Review. *ChemElectroChem* **2024**, *11* (14), No. e202300702.

(20) Huo, H.; Janek, J. Silicon as Emerging Anode in Solid-State Batteries. *ACS Energy Lett.* **2022**, *7* (11), 4005−4016.

(21) Yamada, A. Hidden Negative Issues and Possible Solutions for Advancing the Development of High-Energy-Density in Lithium Batteries: A Review. *Adv. Sci.* **2024**, *11* (25), No. 2401739.

(22) Boorboor Ajdari, F.; Asghari, P.; Molaei Aghdam, A.; Abbasi, F.; Rao, R. P.; Abbasi, A.; Ghasemi, F.; Ramakrishna, S.; Mikaeili Chahartagh, N. Silicon Solid-State Battery: The Solid-State Compatibility, Particle Size, and Carbon Compositing for High Energy Density. *Adv. Funct. Mater.* **2024**, *34* (30), No. 2314822.

(23) Zhang, S.; Yang, Z.; Lu, Y.; Xie, W.; Yan, Z.; Chen, J. Insights into Cation Migration and Intermixing in Advanced Cathode Materials for Lithium-Ion Batteries. *Adv. Energy Mater.* **2024**, *14* (36), No. 2402068.

(24) Xiang, J.; Wei, Y.; Zhong, Y.; Yang, Y.; Cheng, H.; Yuan, L.; Xu, H.; Huang, Y. Building Practical High-Voltage Cathode Materials for Lithium-Ion Batteries. *Adv. Mater.* **2022**, *34* (52), No. 2200912.

(25) Huo, S.; Sheng, L.; Xue, W.; Wang, L.; Xu, H.; Zhang, H.; Su, B.; Lyu, M.; He, X. Challenges of Stable Ion Pathways in Cathode Electrode for All-Solid-State Lithium Batteries: A Review. *Adv. Energy Mater.* **2023**, *13* (15), No. 2204343.

(26) Jeong, W.; Yun, J.; Lee, J.-W. Interfacial Challenges and Recent Advances of Solid-State Lithium Metal Batteries. *Bull. Korean Chem. Soc.* **2024**, *45* (10), 806−820.

(27) Kang, J.; Shin, H. R.; Yun, J.; Kim, S.; Kim, B.; Lee, K.; Lim, Y.; Lee, J.-W. Chemo-Mechanical Failure of Solid Composite Cathodes Accelerated by High-Strain Anodes in All-Solid-State Batteries. *Energy Storage Mater.* **2023**, *63*, No. 103049.

(28) Chen, X.; He, W.; Ding, L.-X.; Wang, S.; Wang, H. Enhancing Interfacial Contact in All Solid State Batteries with a Cathode-Supported Solid Electrolyte Membrane Framework. *Energy Environ. Sci.* **2019**, *12* (4), 938−944.

(29) Kim, J.; Kim, M. J.; Kim, J.; Lee, J. W.; Park, J.; Wang, S. E.; Lee, S.; Kang, Y. C.; Paik, U.; Jung, D. S.; Song, T. High-Performance All-Solid-State Batteries Enabled by Intimate Interfacial Contact Between the Cathode and Sulfide-Based Solid Electrolytes. *Adv. Funct. Mater.* **2023**, *33* (12), No. 221135.

(30) Jung, J. Y.; Kim, K.; Suh, J. H.; Kim, H.-S.; You, M. J.; Park, K.-H.; Song, J. H.; Yu, J.-S.; Lee, J.-W.; Cho, W.; Park, M.-S. Customizing the Morphology and Microstructure of Single-Crystalline Ni-Rich Layered Cathode Materials for All-Solid-State Batteries. *Chem. Eng. J.* **2023**, *470*, No. 144381.

(31) Yang, X.; Doyle-Davis, K.; Gao, X.; Sun, X. Recent Progress and Perspectives on Designing High-Performance Thick Electrodes for All-Solid-State Lithium Batteries. *eTransportation* **2022**, *11* (2), No. 100152.

(32) Kato, Y.; Shiotani, S.; Morita, K.; Suzuki, K.; Hirayama, M.; Kanno, R. All-Solid-State Batteries with Thick Electrode Configurations. *J. Phys. Chem. Lett.* **2018**, *9* (3), 607−613.

(33) Kuang, Y.; Chen, C.; Kirsch, D.; Hu, L. Thick Electrode Batteries: Principles, Opportunities, and Challenges. *Adv. Energy Mater.* **2019**, *9* (33), No. 1901457.

(34) Wang, Y.; Hao, H.; Naik, K. G.; Vishnugopi, B. S.; Fincher, C. D.; Yan, Q.; Raj, V.; Celio, H.; Yang, G.; Fang, H.; Chiang, Y.-M.; Perras, F. A.; Jena, P.; Watt, J.; Mukherjee, P. P.; Mitlin, D. Mechanical Milling-Induced Microstructure Changes in Argyrodite LPSCl Solid-State Electrolyte Critically Affect Electrochemical Stability. *Adv. Energy Mater.* **2024**, *14* (23), No. 2304520.

(35) Shi, T.; Tu, Q.; Tian, Y.; Xiao, Y.; Miara, L. J.; Kononova, O.; Ceder, G. High Active Material Loading in All-Solid-State Battery Electrode via Particle Size Optimization. *Adv. Energy Mater.* **2020**, *10* (1), No. 1902881.

(36) Strauss, F.; Bartsch, T.; Biasi, L.; Kim, A.-Y.; Janek, J.; Hartmann, P.; Brezesinski, T. Impact of Cathode Material Particle Size on the Capacity of Bulk-Type All-Solid-State Batteries. *ACS Energy Lett.* **2018**, *3* (4), 992−996.

(37) Zhao, J.; Zhao, C.; Zhu, J.; Liu, X.; Yao, J.; Wang, B.; Dai, Q.; Wang, Z.; Chen, J.; Jia, P.; Li, Y.; Harris, S. J.; Yang, Y.; Tang, Y.; Zhang, L.; Ding, F.; Huang, J. Size-Dependent Chemomechanical Failure of Sulfide Solid Electrolyte Particles During Electrochemical Reaction with Lithium. *Nano Lett.* **2022**, *22* (1), 411−418.

(38) Schlautmann, E.; Weiß, A.; Maus, O.; Ketter, L.; Rana, M.; Puls, S.; Nickel, V.; Gabbey, C.; Hartnig, C.; Bielefeld, A.; Zeier, W. G. Impact of the Solid Electrolyte Particle Size Distribution in Sulfide-

Based Solid-State Battery Composites. *Adv. Energy Mater.* **2023**, *13* (41), No. 2302309.

(39) Han, S. C.; Ali, M.; Kim, Y. J.; Park, J.-H.; Lee, Y.-J.; Park, J.-W.; Park, H.; Park, G.; Lee, E.; Kim, B. G.; Ha, Y.-C. Unraveling Electrochemo-Mechanical Aspects of Core−Shell Composite Cathode for Sulfide-Based All-Solid-State Batteries. *J. Mater. Chem. A* **2024**, *12* (37), 24896−24905.

(40) Wang, C.-W.; Yi, Y.-B.; Sastry, A. M.; Shim, J.; Striebel, K. A. Particle Compression and Conductivity in Li-Ion Anodes with Graphite Additives. *J. Electrochem. Soc.* **2004**, *151* (9), No. A1489.

(41) Klumov, B. A.; Khrapak, S. A.; Morfill, G. E. Structural Properties of Dense Hard Sphere Packings. *Phys. Rev. B* **2011**, *83* (17), No. 179903.

(42) Koerver, R.; Zhang, W.; Biasi, L.; Schweidler, S.; Kondrakov, A. O.; Kolling, S.; Brezesinski, T.; Hartmann, P.; Zeier, W. G.; Janek, J. Chemo-Mechanical Expansion of Lithium Electrode Materials − On the Route to Mechanically Optimized All-Solid-State Batteries. *Energy Environ. Sci.* **2018**, *11* (8), 2142−2158.

(43) Oh, J.; Chung, W. J.; Jung, S. H.; Kim, Y.; Lee, Y.; Nam, Y. J.; Lee, S.; Kim, C. H.; Cho, J. W. Critical Impact of Volume Changes in Sulfide-Based All-Solid-State Batteries Operating Under Practical Conditions. *Energy Storage Mater.* **2024**, *71*, No. 103606.

(44) Su, Y.; Zhang, Q.; Chen, L.; Bao, L.; Lu, Y.; Chen, S.; Wu, F. Stress Accumulation in Ni-Rich Layered Oxide Cathodes: Origin, Impact, and Resolution. *J. Energy Chem.* **2022**, *65* (2), 236−253.

(45) de Biasi, L.; Schwarz, B.; Brezesinski, T.; Hartmann, P.; Janek, J.; Ehrenberg, H. Chemical, Structural, and Electronic Aspects of Formation and Degradation Behavior on Different Length Scales of Ni-Rich NCM and Li-Rich HE-NCM Cathode Materials in Li-Ion Batteries. *Adv. Mater.* **2019**, *31* (26), No. 1900985.

(46) Ito, S.; Fujiki, S.; Yamada, T.; Aihara, Y.; Park, Y.; Kim, T. Y.; Baek, S.-W.; Lee, J.-M.; Doo, S.; Machida, N. A Rocking Chair Type All-Solid-State Lithium Ion Battery Adopting Li_2O-ZrO_2 Coated $LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ and a Sulfide Based Electrolyte. *J. Power Sources* **2014**, *248*, 943−950.

(47) Cruz-Manzo, S.; Greenwood, P. An Impedance Model Based on a Transmission Line Circuit and a Frequency Dispersion Warburg Component for the Study of EIS in Li-Ion Batteries. *J. Electroanal. Chem.* **2020**, *871*, No. 114305.

(48) Shen, W.; Huang, J.; Qu, X.; Li, K.; Fu, G.; Chen, X.; Lin, Y. Cracking Mechanism and Inhibition Strategies of Polycrystalline NCM Electrode Particles. *ACS Appl. Mater. Interfaces* **2024**, *16* (42), 57074−57090.

(49) Trevisanello, E.; Ruess, R.; Conforto, G.; Richter, F. H.; Janek, J. Polycrystalline and Single Crystalline NCM Cathode Materials—Quantifying Particle Cracking, Active Surface Area, and Lithium Diffusion. *Adv. Energy Mater.* **2021**, *11* (18), No. 2003400.

(50) Fathiannasab, H.; Zhu, L.; Chen, Z. Chemo-Mechanical Modeling of Stress Evolution in All-Solid-State Lithium-Ion Batteries Using Synchrotron Transmission X-Ray Microscopy Tomography. *J. Power Sources* **2021**, *483*, No. 229028.

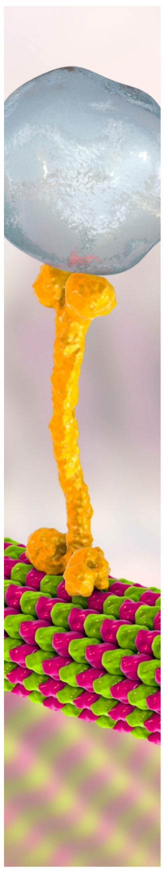
(51) Yu, H.-C.; Taha, D.; Thompson, T.; Taylor, N. J.; Drews, A.; Sakamoto, J.; Thornton, K. Deformation and Stresses in Solid-State Composite Battery Cathodes. *J. Power Sources* **2019**, *440*, No. 227116.

(52) Liu, X.; Zheng, B.; Zhao, J.; Zhao, W.; Liang, Z.; Su, Y.; Xie, C.; Zhou, K.; Xiang, Y.; Zhu, J.; Wang, H.; Zhong, G.; Gong, Z.; Huang, J.; Yang, Y. Electrochemo-Mechanical Effects on Structural Integrity of Ni-Rich Cathodes with Different Microstructures in All-Solid-State Batteries. *Adv. Energy Mater.* **2021**, *11* (8), No. 2003583.

(53) Walther, F.; Koerver, R.; Fuchs, T.; Ohno, S.; Sann, J.; Rohnke, M.; Zeier, W. G.; Janek, J. Visualization of the Interfacial Decomposition of Composite Cathodes in Argyrodite-Based All-Solid-State Batteries Using Time-of-Flight Secondary-Ion Mass Spectrometry. *Chem. Mater.* **2019**, *31* (10), 3745−3755.

(54) Sun, X.; Wang, L.; Ma, J.; Yu, X.; Zhang, S.; Zhou, X.; Cui, G. A Bifunctional Chemomechanics Strategy to Suppress Electrochemo-Mechanical Failure of Ni-Rich Cathodes for All-Solid-State Lithium Batteries. *ACS Appl. Mater. Interfaces* **2022**, *14* (15), 17674−17681.

(55) Li, Y.; Zhang, D.; Xu, X.; Wang, Z.; Liu, Z.; Shen, J.; Liu, J.; Zhu, M. Interface Engineering for Composite Cathodes in Sulfide-Based All-Solid-State Lithium Batteries. *J. Energy Chem.* **2021**, *60* (9), 32−60.



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